What Decision Records Establish About Model Choices Under Uncertainty

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Abstract

Decision records can be evaluated by whether they match an action, identify an executable selection rule, or help predict a later choice. We distinguish these questions through a case study of model decisions under uncertainty. In a concrete counterexample, five ways of completing an incompletely specified selector reproduce the same original action: one plan is best on all five criteria. Applied to another fixed representation, those selectors yield different preservation/loss labels under the study's conditional feasibility evaluation. Original-action agreement therefore leaves the resulting component diagnosis ambiguous. In a separate follow-up, required records match all sixteen model proposals executed without external selection. The two supported rules agree in all sixteen records, with fourteen agreements forced by a single row and two observed in multirow tables. Finally, a prospective pilot freezes eight new records before sixteen target choices across two execution segments. Prediction with versus without the record is 6/8 versus 7/8 for Astra and 3/8 versus 3/8 for Sol: no positive net accuracy gain is observed across four purposeful histories per profile. The study provides concrete controls for evaluating decision records: test whether candidate rules can disagree, separate action correspondence from task labels, and assess added prediction on later choices. These measurements concern external records in two task families; population effects require broader sampling.

1 Question and contribution

Our Research Under Uncertainty (RUU) program studies how models form assumptions, choose experiments and revise predictions when an environment's mechanism is unknown. Decision records offer a way to examine those activities. Before using them as evidence, however, we need to know which relationships they measure. This paper asks: **when does a record that matches a model's action identify a selection rule, and when does it help predict a later choice?**

We study externally supplied tables and descriptions in two simulated task families. An executable record combines an action-scoring representation with a deterministic rule for selecting an action. For an incompletely specified record, the analyst must supply the remaining choices; we call this a *selector completion*. Combining one record's representation with another record's rule is a *crossing*. For example, applying column means and column maxima to the same table can select different actions. Our initial component diagnostic used such crossings to examine which part of a record supported a successful choice.

A concrete identification problem. In the pilot's original representation A2, one plan is best on all five examined criteria. Total cost and trajectory peak therefore reproduce the same model action. In a second representation A1, held fixed for comparison, the criteria separate: total cost selects a plan with a preservation witness, while trajectory peak selects a different plan with a conditional loss certificate. Agreement on A2 leaves these completions indistinguishable, although they give different crossed diagnoses (Section 4.1). This example supplies a practical check: *does the validation situation give candidate selectors*

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an opportunity to disagree? When it does not, the diagnosis must remain conditional on the independently specified selection contract.

Correspondence and prediction are separately measurable. A follow-up separates requiring a record from externally selecting its action. Each profile supplies eight records matching its own executed proposal. Fourteen of these sixteen numeric tables have one row, so both supported rules necessarily agree; their agreement in the remaining two multirow tables is observed. We pair these counts with an action-copy control and a scan over the supported rules (Section 5). This makes both correspondence and its opportunity for rule discrimination visible.

A prospective pilot then tests added prediction on new choices. Records are frozen before two target branches per source, and paired observers receive the same information with or without the record. The observed net accuracy differences are $-1/8$ for Astra and $0/8$ for Sol across four purposeful histories per profile (Section 6). The pilot measures transfer from a source that reports an action and record to a target that reports an action alone; it evaluates the numeric table and accompanying prose together.

Contribution to evaluation practice. The evaluation decision at issue is accepting an analyst-completed selector because it reproduces the source action. The archived A2 case exposes the problem: one plan is best on all five criteria, so every tested completion passes that check, yet the same completions produce different choices and conditional task diagnoses on A1. Researchers using such records should therefore retain the compatible selector set and test a situation where its members can disagree before attributing a diagnosis to a particular rule.

The follow-up and prospective panels make two further reporting decisions concrete: separate a record's match to an unforced proposal from the task label, and measure added prediction on new choices after freezing the record. An action change can preserve its P or L label (Table 11); an aggregate prediction difference can combine opposite observed contrasts across tasks (Section 6). These cases give researchers specific checks to add to a record evaluation. The common-menu recovery identity and single-row rule agreement explain what the contract already guarantees (Appendix K); the archived cases and model decisions show where the distinctions matter in practice. Table 1 maps each panel to its question, and Section 7 relates these external-record measurements to explanation evaluation.

Study	Question	Intervention or check	Supported claim
Pilot (8 trials)	Does native agreement identify a selector?	Five analyst completions of archived records; component crossings	Concrete non-identifiability under these completions
Transfer (48 trials)	Are free-form records crossing-ready?	Optional records across two task families	Applicability is limited; zero eligible pairs is not success
Common menu (32 trials)	What does the fixed numeric contract add?	Optional records versus required records with external selection	Structural recoverability; record requirement and execution remain combined
Follow-up (48 assignments)	Can record and proposal match without forced execution?	A optional/model, B required/model, C required/selector	Profile-specific correspondence, record structure and conditional labels; no conflicting C proposal observed
Prospective choice prediction (16 targets)	Does a frozen record add prediction value?	Paired observers with/without record; same public evidence and profile	No positive net accuracy gain observed; four histories/profile, two linked targets/history

Table 1: Study map. Counts refer to distinct panels, not pooled replications. The follow-up has 46 completed assignments and two retained technical losses. The prospective pilot has 15 original targets plus one separately completed target; its original interrupted request remains recorded. The separate finite-class companion is not needed for the main argument.

2 Task, record, execution and evaluation

Task. The actor receives an action/observation history, a remaining objective and a finite menu, without the hidden mechanism. In the continuous task it chooses a bounded two-dimensional control to preserve the possibility of reaching and maintaining a target region under a path constraint. In the route task it chooses a legal discrete action under remaining route constraints. The common-menu and prospective panels offer 13 continuous controls or three route actions, and allow actors up to three analyses of the observed history. Both families supply the past actions and observed states, the remaining objective and constraints; the hidden transition mechanism is withheld. The prospective panel's concrete source–target changes appear in Table 21.

Record. The actor can submit a numeric table of candidate failure forecasts and a declared rule. Columns identify menu actions; rows contain values in $[0, 1]$. Uniform mean minimizes the column mean; worst case minimizes the column maximum; ties use the lowest index. The executable contract specifies action selection; forecast calibration and the semantic adequacy of rows require separate evaluation. Table 2 shows one complete numeric table from the follow-up, before any formal crossing notation is needed.

Archived candidate-loss row	Menu action 0	Menu action 1	Menu action 2
Only row	0.35	0.30	0.25

Table 2: A real B record: Astra, route history `probe-03`, repeat 1, trial 3. The declared mean rule selects action 2, as does worst case. The model proposed action 2 and B executed that proposal. Its bound menu label is P. The numbers are uncalibrated forecasts; their ordering and the matching proposal are reproducible, but their semantic adequacy or causal role is not established. Exact final JSON and identity bindings are in the bundle.

Execution. The follow-up separates record requirement from action source (Table 3). In B the model proposal is executed even if a valid record would select a different action. In C the external selector determines execution, while the proposal remains observable. Thus B's record–proposal agreement is observed; C's record–execution agreement is guaranteed. Exact instructions and schema are in the bundle.

Arm	Record requirement	Executed action	What agreement can establish
A	Optional	Model proposal	Correspondence, if a valid record exists
B	Required valid record	Model proposal	Observed record–proposal correspondence
C	Required valid record	Declared record selector	Execution match by construction; proposal match remains observable

Table 3: Follow-up conditions. A valid record need not match the proposal to be accepted in B. A missing record leaves correspondence undefined.

Evaluation. The analyst attaches a task label to the executed menu action: preservation (P), loss (L), or unresolved (U), using the bound feasibility evaluation. That label is separate from whether a record is valid, matches the proposal, or changes action under rule substitution. The model does not receive the privileged evaluator during collection.

2.1 Why matching needs a control

For any proposed index a in an n-action menu, construct one row $A_{1j} = 0$ when $j = a$, and $A_{1j} = 1$ otherwise. Both supported rules select a. This *action-copy negative control* requires only the menu size and proposed index, with no history, mechanism or task label. Conditional on the proposal it supplies no independent information about how that proposal was produced. It nevertheless satisfies validity and correspondence checks. Consequently those checks alone cannot establish an informative explanation. The construction

tests the sufficiency of those checks. How the participant produced its record and whether its forecasts are informative remain empirical questions.

A positive arithmetic control has two rows, $A = \left(\begin{smallmatrix} 0 & 0.4 \\ 0.7 & 0.4 \end{smallmatrix}\right)$. Mean scores $(0.35, 0.4)$ select action 0; worst-case scores $(0.7, 0.4)$ select action 1. The same reader therefore detects a rule contrast when the values support one. Neither constructed control has a physical task label or constitutes an additional model trial. The distributed check tests the copy construction at every proposal index for menu sizes 3 and 13, under each declared rule, and tests the positive control under both rules.

2.2 Two external comparisons

Let A_i be record i's table over a common finite menu and let r_i be its declared deterministic rule. The action $a_i = r_i(A_i)$ is one column of that menu. Here, a_i denotes the record-selected action; the model-proposed and executed actions are separate quantities. The *observed-rule crossing* has four entries $c_{ij} = r_j(A_i)$ for $i, j \in \{1, 2\}$. The separate *supported-rule-family table* evaluates each A_i under both allowed rules: uniform mean and worst case, with lowest-index tie breaking. The family table may contain a rule that neither paired actor used. Its entries are deterministic evaluations of archived records, with task labels from the bound menu census; a live model counterfactual would require a new model decision.

2.3 Feasibility labels and access

P requires a successful admissible continuation witness after the action. L requires the specified loss certificate; absence of a witness alone is not L. U preserves unresolved numerical cases. Continuous-control labels are conditional on the disclosed floating-point and bounding premises in the supplement. The discrete route family's finite computation has a different verification contract. Both actors and analyst see the legal histories; only the analyst receives the hidden mechanisms and privileged evaluator during the original experiment. Reviewer disclosure now permits inspection of these mechanisms without retroactively changing what participants received.

Record validity, correspondence and rule sensitivity can be checked without the physical label map. P/L/U comparisons add its assumptions. The exact arithmetic helper does not prove the uniform library bounds or attest the historical runtime; those premises remain open. These are distinct evidence levels when interpreting outcome differences.

3 Evidence sets and their denominators

The first three evidence sets below use crossing-pair denominators. The separate follow-up in Section 5 instead compares A/B/C conditions within eight matched triplets per profile; these condition pairs are not crossing pairs. Its 48 assignments include two technical losses.

Evidence set	Trials	Pairs	Eligible records	Eligible pairs
Pilot (component study)	8	4	2 selected component pairs	
Prospective transfer	48	24	1	0
Optional record	16	8	1	0
Mandatory record + execution	16	8	16	8

Table 4: The three predecessor evidence sets. Pilot component eligibility is a different construction from the frozen prospective adapter and common-menu contract. Empty pair denominators leave the crossing effect unassessed.

The pilot used four fixed histories with two repeats under the requested gpt-6-astra/max profile. Its component study focused on two boundary histories, pilot contexts B and D, after outcomes were available. These history names are unrelated to the follow-up arms A/B/C. The transfer panel used 48 live trials: two families, three instances per family, two histories per instance, the gpt-6-astra/max and gpt-6.1-sol/max profiles,

and two repeats. Under its frozen adapter, a record had to support the prescribed operation without post hoc completion. Only one trial qualified, leaving zero of 24 pairs crossable. This measures the adapter's applicability to the available external records. With no eligible pair, the crossing effect is unassessed; internal representations are outside the adapter's observations. Native transfer labels were 42 P and 6 L. The adapter's mutually exclusive outcomes were: 24 route trials with no analysis attempt, 18 whose last successful analysis lacked a table, three whose rule did not reproduce the action, two with a menu-identity mismatch, and one supported record. These reasons distinguish absent evidence from an invalid submitted record. The other 24, continuous-task trials made 72 analysis attempts: 67 succeeded and five returned errors. The accounting distinguishes an unused analysis opportunity from a failed analysis call. It measures visible tool use, leaving internal reasoning outside the observation.

The 32-trial panel crossed two task families, two purposefully selected histories per family, two model profiles (gpt-6-astra/max and gpt-6.1-sol/max), two repeats and two conditions. Each condition has 16 trials and eight pairs. A crossing pair comprises the two repeats at the same history and profile within one condition, not a pair of trials from different conditions. Both conditions used a common finite menu and structured final output. Appendix G identifies the selected histories, execution contract, resource limits and recorded usage. The optional condition permitted an observational record; the mandatory condition required a record and let an external deterministic executor select the action. Mandatory record–action agreement is consequently a construction property, unlike a matching optional observational record. The combined intervention does not isolate a pure reporting effect.

All 32 trials were evaluated: no unstarted slots, technical missingness or protocol-invalid trials remain. This does not mean the analysis interface was error-free: 48 attempts yielded 41 successes and seven code errors in seven trials (five optional, two mandatory). Five errors used attribute access on the dictionary history; two used the unavailable built-in `next`. The prompt described `history.states/history.actions`, although the worker passed a dictionary, and did not enumerate the built-ins. Each affected trial consumed all three analysis slots, one on the error and two successfully. Appendix I gives the trial-level accounting with all original trials and labels retained. The effect on subsequent actions and record structure is unassessed. Optional P/L/U counts were 13/3/0; mandatory counts were 10/5/1. Four purposefully chosen histories do not form a population sample of tasks or support a model ranking. Table 5 separates the families' label contracts. The study inventory in Appendix H identifies excluded historical and development work; the later acquisition experiment has its own denominator.

Condition	Family	P	L	U	Verification contract
Optional	Continuous	5	3	0	Numerical, conditional
Mandatory	Continuous	2	5	1	Numerical, conditional
Optional	Route	8	0	0	Finite integer recurrence
Mandatory	Route	8	0	0	Finite integer recurrence

Table 5: Native outcomes in the common-menu panel, eight trials per row. Continuous P uses an admissible continuation witness; continuous L depends on the numerical and historical-runtime premises in Appendix F. U remains unresolved. Successful present-host replay does not close those premises.

4 Results: what agreement and crossing identify

4.1 Pilot counterexample: native agreement does not identify the selector

The missing choice can be seen in pilot context B. Its two archived model analyses supplied different candidate dynamics and planning ingredients; the native-loss representation A1 has eight candidates, while the native-preservation representation A2 has four. The researcher reconstruction computes one multi-step plan per candidate. To execute this component on either representation, an analyst must also specify *which plan supplies the next action*. The declared completion R2* chooses the plan with the smallest total penalized cost. This last selection is an analyst choice (Appendix C), not identified by reproducing the native action.

Here *native-action-compatible* means only that the selected first action on A2 is within 10^{-6} of the recorded native action. It does not mean agreement with every semantic commitment of the model's narrative or recovery of its internal policy. We examine five selectors over the same fixed plans: total cost, maximum target-step norm, mean squared target-step norm, trajectory peak norm and final-step norm. These are explicit summaries of the plans' objective, target attainment or path behavior, making the analyst's choice inspectable. They obey the same action bounds and lowest-index tie breaking, but need not encode the same preferences; final-step norm alone, for example, is not the full multi-step task criterion. This is a finite retrospective sensitivity family, not an exhaustive set of faithful completions.

Criterion minimized	Native A2: plan/label	Crossed A1: plan/label	Action norm
Total penalized cost (R2*)	2/P	2/P	0.575465
Maximum target-step norm	2/P	2/P	0.575465
Mean squared target-step norm	2/P	2/P	0.575465
Trajectory peak norm	2/P	3/L	0.042684
Final-step norm	2/P	5/P	0.579114

Table 6: Pilot context B: five selectors applied to the same fixed plan pools. Indices are zero-based. All five select native A2 plan 2 and reproduce the original action within 10^{-6}. The first three select the same crossed A1 plan; the minimum-peak selector instead selects plan 3, giving L rather than P. Norms refer to crossed actions. Labels retain the conditional continuous evaluation premises. These five selectors are not independent replications or a probability sample of completions.

All five select A2 plan 2, whose first action matches the native action to approximately 1.81×10^{-9}. Plan 2 has a strictly lower value than each of the other three plans on every one of the five criteria. A2 thus provides no opportunity to discriminate among these selectors: agreement there follows from a common best plan. Now hold the crossed A1 representation, history and computed plans fixed. Total cost selects A1 plan 2, with action norm 0.575465 and a preservation witness. Trajectory peak instead selects A1 plan 3, with norm 0.042684 and a conditional loss certificate (Table 6). The reader command recomputes all twelve plans and five selections, re-evaluates their first actions with the bundled grader, and compares the complete archived witness/certificate records (Section 9). It therefore checks the label change as well as the selector arithmetic. The numerical and historical-runtime premises in Appendix F remain open.

The resulting component diagnosis depends on which native-compatible selector the analyst supplies. To use native-action agreement as a validation test, report the candidate selectors and whether the validation cases actually separate their choices. An independently specified selection contract can restrict the alternatives. In this case, the common best plan on A2 leaves that choice open. The two A1 plans also differ in action magnitude, so this contrast evaluates the full selection change rather than isolating a horizon or direction effect.

Under the originally declared completions, pilot context B had two P off-diagonals and context D two L off-diagonals. Each native pair already contained one P and one L. Those patterns describe completion-conditional component dependence, not new outcome categories or internal causal ground truth. The pilot's heterogeneous optimizers differ from the later common-menu contract; the latter's structural identity is not extended to this setting.

4.2 Common-menu applicability and recoverability

The mandatory condition made all eight pairs executable. It did not make their crossing tables informative by construction. Every pair used the same declared rule, so all 32 observed-rule cells were already determined by native actions and rule identity. Among the 16 mandatory records, 13 had one row, one contained duplicate rows alongside distinct rows, and 13 had only one distinct row. A duplicate-containing table can also contain distinct rows, so the duplicate and single-distinct-row counts measure different properties.

Native actions plus the single-distinct-row check determine 29 of the 32 supported-rule-family cells. The remaining sensitivity can be calculated on each record independently. Appendix K states the fixed-contract

identity and the complete recoverability table. These calculations measure recoverable cells; improved diagnostic accuracy requires an external diagnostic target.

Fifteen of 16 mandatory records selected the same action under both supported rules. The remaining record changed action and its conditional continuous P/L category; the independent scan recovered this change. One pair involved U and remains unresolved rather than being converted into a resolved loss. The two 32-cell displays overlap; after within-pair deduplication they reference 30 unique numerical computations, not 64 new samples. Cross-action labels reuse the existing bound menu census. No new live model counterfactuals were collected for this retrospective analysis.

Worked same-record contrast. For the mandatory `probe-01` record from gpt-6-astra, repeat 1, the independent scan yields:

Rule applied to this record	Menu index	Bound task label
Uniform mean	7	L (conditional continuous certificate)
Worst case	12	P (continuation witness)

Indices are zero-based. The recorded table, menu identity and label census are held fixed. No partner record is needed to expose this sensitivity; it concerns the declared external rule, not a demonstrated internal cause. The alternative action's label is a deterministic menu evaluation, not an additional live model decision. This is the sole rule-sensitive mandatory record, so it must not be presented as a prevalent population effect. This trial was also affected by the dictionary-access error. The archived table's rule sensitivity is reproducible; whether correcting the interface would have changed that table or the decision is unknown.

The optional condition contains one submitted record, which is eligible, and 15 trials with no submitted record (`no_record`). It has no eligible pairs. The fifteen absences and one eligible submission have different statuses. With no eligible pair, the optional crossing effect remains unassessed. The panel thus locates an applicability bottleneck in the optional condition and a redundancy bottleneck under mandatory recording with external selection.

5 Follow-up: required records with and without external execution

The earlier mandatory condition combined a record requirement with an external executor. The follow-up adds a required-record condition that executes the model's proposal. Each matched triplet uses one history, repeat and model/max/Fast profile: A permits an optional record and executes the proposal; B requires a valid record and executes the proposal; C requires a valid record and executes its declared selector, retaining the proposal for observation. Every trial starts a fresh context. Four purposefully selected historical contexts, two profiles and two repeats give 48 planned assignments, eight triplets per profile. These repeats are not independent draws of new worlds. The allocation and six possible within-triplet orders were fixed before collection; earlier-panel trials are not reused as controls. Appendix J gives the execution segments, interface and reader boundaries. No new population sample or live crossed decision was collected.

Availability and correspondence. In A, no valid record was available among Astra's seven or Sol's eight completed trials. This is absence of the specified external record, not absence of reasoning. In B, each profile supplied eight valid records and all eight matched its own proposal. Unlike C's executed-action agreement, this correspondence was not imposed by the executor. It does not establish that the record caused the proposal or faithfully represents internal reasoning. Table 8 separates the observed record structures from their mathematical consequences. Astra supplied eight single-row B records; Sol supplied six single-row and two multirow B records. For every single-row record, equality of mean and worst case is forced, so it is not an independent empirical success of rule agreement. Sol's multirow records contain three and four distinct rows; both also select the same action under the two rules. That equality is an observation about these values, not forced by having one distinct row. Their source orders are 44 and 47, respectively. No multirow conclusion is available for Astra's empty denominator.

Profile	Arm	Done	Tech.	P/L/U	Valid	Match	One row	Same action
Astra	A	7	1	6/1/0	0	–	0	0
Astra	B	8	0	8/0/0	8	8/8	8	8
Astra	C	7	1	5/2/0	7	7/7	6	6
Sol	A	8	0	7/1/0	0	–	0	0
Sol	B	8	0	6/2/0	8	8/8	6	8
Sol	C	8	0	7/1/0	8	8/8	7	7

Table 7: Follow-up, eight planned assignments per row; profiles are separate strata, not ranked. Done means a completed trial, not full task success. Valid counts usable records; Match is record-selected action versus model proposal among valid records. The empty A denominator is undefined. Same action counts records for which both supported rules select the same action. Sol C has one duplicate-containing record; all other rows have zero. There are no protocol losses, active or unstarted assignments.

Profile	Proposal matches	One row	Multiple rows	Multirow same action
Astra	8/8	8	0	–
Sol	8/8	6	2	2/2

Table 8: B records, separated by profile. One-row mean/worst-case equality is forced by the contract. For multirow records, equality is an observed property of the archived values. Astra has no multirow denominator, so its dash is undefined, not a success. Sol's two multirow records have three and four distinct rows, respectively.

The empirical result is the co-occurrence of proposal matching and these record structures without external selection. The action-copy control in Section 2 establishes why matching alone leaves substantive use of the history unresolved. The actual records' predictive value requires its own measurement. Single-row and duplicate records satisfy the protocol and remain in the structural accounting. For evaluation practice, a correspondence rate should therefore be accompanied by the number of distinct rows, the actual choices under alternative supported rules, and the action-copy control. These checks separate an observed match from an opportunity to discriminate between rules; they do not replace a test of prediction on unseen decisions.

Conditional outcomes and observed opportunities. Table 9 reports the two predeclared contrasts using only observed nontechnical pairs. Astra has seven fully completed triplets and Sol eight. L outcomes remain in the comparison; technical failures are reported separately and are not converted into L. The directions do not support a common improvement claim across these profiles, nor an equivalence or no-effect conclusion. In Sol B, eight matching records coexist with two L labels, illustrating why correspondence and continuation feasibility require separate measurements. P certifies a conditional feasible continuation; the experiments do not measure completed-task success.

Profile	Contrast	Pairs	Missing	L→P	P→L	Same	Net P
Astra	B–A	7/8	1	1	0	6	+1
Astra	C–B	7/8	1	0	2	5	-2
Sol	B–A	8/8	0	0	1	7	-1
Sol	C–B	8/8	0	2	1	5	+1

Table 9: Descriptive paired transitions run from A to B for B–A, and from B to C for C–B. Pairs are observed/planned; Missing counts technical missing pairs; Same includes P-to-P and L-to-L. Dividing Net P by observed pairs recovers the reported paired difference. No pooled estimate or population inference. C–B is the total condition contrast, not an isolated executor effect on the same fixed record.

Where the outcome contrasts occur. The family breakdown in Table 10 locates every observed P/L movement in the two continuous-control histories. All completed route decisions were P, although each route

menu contained an L option; this pattern was not forced by having only P choices. The continuous menus have 13 actions and the route menus three, so these purposeful contexts are not a calibrated difficulty ladder. Appendix J.1 reports every profile/history/arm cell. In particular, Sol's C–B net gain of one P comprises two L-to-P and one P-to-L transitions. Net change alone does not describe those opposing movements, and the underlying continuous labels retain the numerical/runtime premises stated above.

Profile	Family	Assigned	Done	Tech.	P/L/U
Astra	Continuous	12	12	0	9/3/0
Astra	Route	12	10	2	10/0/0
Sol	Continuous	12	12	0	8/4/0
Sol	Route	12	12	0	12/0/0

Table 10: Follow-up outcomes by family, separately within each profile. Each family has two purposeful histories and two repeats per arm. Done denotes a completed decision, not full task success. P/L labels in the continuous family retain the numerical/runtime premises.

A changed action need not change the task label. The follow-up contains two rule-sensitive records, both in C (Table 11). Mean and worst case select different menu indices, while the bound label remains P in one and L in the other. These are existing-record calculations, not new live counterfactual trials. Together with the older panel's single label-changing example, they make action sensitivity and task-label sensitivity separately inspectable.

Profile	History	Repeat	Arm	Mean	Worst case
Astra	probe-01	2	C	1 (P)	0 (P)
Sol	probe-02	1	C	11 (L)	0 (L)

Table 11: The two rule-sensitive follow-up records. Entries give zero-based menu index and bound label. The action changes but the label does not; these are per-record calculations on existing menus, not live crossed decisions. Continuous labels remain conditional.

Every completed C proposal already matched the record-selected action (Astra 7/7, Sol 8/8). No observed C trial therefore tests the benefit of overriding a conflicting proposal. The C–B contrast is not an isolated mechanical correction effect on the same fixed record. Labels and within-record rule diagnostics are tied to the existing menu contract; they are not additional live crossing observations or independent evidence about new task physics.

6 Prospective pilot: does a frozen record predict a new choice?

We test whether a source decision record improves prediction of later model choices beyond the public history and source action. Four new purposefully constructed histories in the two disclosed task families are used separately for Astra/max/Fast and Sol/max/Fast. Each source has two precommitted target branches: eight target choices per profile, grouped in four histories. The histories, records and target choices form a new panel. Each target extends its source history by one observed transition under a fixed exogenous action: continuous sources at step 6 branch to step 7 under $(0.6, 0)$ or $(-0.6, 0)$; route sources at step 4 branch to step 5 by advancing one or three nodes. These alternatives were set before source-model outputs and need not follow the source proposal. Table 21 and Appendix L give the construction and selection basis.

Information and output protocols. The source actor's final response contains an action, a numeric table with a declared rule, and four brief fields for assumptions, uncertainty, criterion and expected observation. The record is frozen before the source actor sees either target. Two fresh-context observers of the same requested profile predict the target actor's top-1 action with or without that record. Both receive the source action, source and target histories, menus, and the complete source and target instructions and output

schemas. Only the record differs between observer inputs. Their order is preallocated, allowances are equal, and both predictions are locked before the fresh target actor runs.

The target's final response contains an action alone. Source and target may both request up to three local analyses before their final response, and their proposals are unforced. The target receives the public evidence and source action, without the record or observer predictions. The comparison therefore measures the record's added value for predicting an action under this target protocol. Both observers predict the same target; the effect of changing the source–target output protocol is not separately identified.

Panel	Profile	Pairs	Record	No record	Copy	Changed	Difference
Initial	Astra	8/8	6/8	7/8	5/8	3/8	-0.125
Initial	Sol	7/8	3/7	3/7	0/7	7/7	$[-0.125, +0.125]$
Combined	Astra	8/8	6/8	7/8	5/8	3/8	-0.125
Combined	Sol	8/8	3/8	3/8	0/8	8/8	0.000

Table 12: Prospective top-1 action prediction. Pairs retains the eight assigned targets/profile; other fractions use available valid targets. Difference is the equally weighted mean over four history clusters, with two target slots each. The initial Sol interval is a missing-data identification bound over all eight assigned targets, not a confidence interval; its observed 3/7 versus 3/7 comparison has difference zero. Copy uses the old source action; Changed compares the target action with that action. Profiles are not ranked.

Profile	Both correct	Record only	No-record only	Both wrong
Astra	5	1	2	0
Sol	2	1	1	4

Table 13: Joint top-1 prediction correctness for the combined panel, eight targets per profile. Record only and No-record only identify which observer is correct on the same target. Each cell uses one generation per role; the counts describe paired outcomes, not case-level causal effects.

Prediction results. The combined panel yields 6/8 versus 7/8 correct predictions for Astra and 3/8 versus 3/8 for Sol, with versus without the record. The mean of the four within-history paired differences is -0.125 for Astra and 0 for Sol. Table 13 shows the joint outcomes: Sol's equal totals include one target correctly predicted only with the record and another correctly predicted only without it. Across these assignments, no positive net accuracy gain is observed. Each cell uses one generation per observer and target role; a discordant pair does not by itself identify a case-level causal benefit or harm.

Where the prediction contrasts occur. Table 14 adds a post hoc descriptive breakdown. Sol's aggregate zero combines one additional correct prediction with the record on continuous tasks and one fewer on route tasks. Astra's one fewer correct prediction occurs in continuous control; both observers predict all four route targets correctly. Each family contains two purposeful histories with two linked targets each. The prespecified per-profile aggregate remains primary; these counts locate the observed contrasts without estimating a family-level causal effect. Table 22 gives all eight history/profile units.

Profile	Family	With record	Without record	Net correct
Astra	Continuous	2/4	3/4	-1
Astra	Route	4/4	4/4	0
Sol	Continuous	3/4	2/4	$+1$
Sol	Route	0/4	1/4	-1

Table 14: Post hoc descriptive family breakdown of the combined prospective panel. Each family has two purposeful histories with two linked targets each. Net correct is the count with minus without the record. The prespecified per-profile aggregate remains primary; these four-target contrasts do not estimate family-level causal effects.

Copying the source action predicts 5/8 Astra and 0/8 Sol targets. The target changes action relative to the source in 3/8 and 8/8 cases, respectively. This baseline describes source–target action change; variability across repeated runs of an identical target input was not measured.

Record structure and interpretation. All four source records per profile match the unforced source proposal. Seven of the eight numeric tables contain a single row: four for Astra and three for Sol. Agreement of the two supported rules is forced in these seven tables and observed in the remaining two-distinct-row Sol table. No duplicate row occurs. The semantic informativeness of the accompanying prose was not independently evaluated. The treatment is the complete record, including its prose; table row counts alone cannot explain the prediction result.

The reference controls verify protocol handling and prediction of supplied hand-authored policies (Appendix L). Sensitivity to an informative record in the scientific tasks remains a separate control target. The observed contrast combines record information, observer use, the source–target protocol change and one draw of each observer and target. It does not separate those explanations for the net result. Task-matched informative-record controls, a same-output-protocol comparison and repeated identical-input decisions would distinguish them in a further study. With four purposeful histories per profile and two linked targets per history, the present pilot characterizes these assignments.

Execution segments. The original run produced fifteen targets. After inspecting its partial results, we completed the remaining assigned Sol block in a separate segment with the same source, inputs, profile and allowances and four fresh reference checks. Table 12 retains both panels. The completion adds one target to an existing history; no completed scientific decision was rerun. Appendix L retains the interrupted attempt, all assigned roles and usage. The two segments may differ in time or provider state; they form one descriptive pilot, not a confirmatory replication.

7 Position relative to existing work

Execution and component diagnosis. Lyu et al. (2023) translate a problem into an intermediate representation and use a deterministic solver to obtain an answer, locating their faithfulness guarantee in the solution stage. Our C condition similarly binds execution to an external record. We examine the additional evaluation questions raised by constructing and comparing such records: selector-completion sensitivity, correspondence without enforced execution, and prediction of later choices. CAR estimates effects of interventions on agent steps through same-policy downstream replay (Shah, 2026); AgentRx uses constraint evidence and a judge for execution diagnosis (Barke et al., 2026); and BLADE aligns evidence from action stages for inspection (Guo et al., 2026). RUU's component diagnoses instead attach feasibility labels to actions selected from external records. The diagnosed object and the outcome measure therefore differ from stochastic-policy replay or failure attribution.

Alternative explanations and discriminating tests. Jain & Wallace (2019) show that substantially different attention distributions can preserve model predictions. Wiegreffe & Pinter (2019) argue that explanation claims depend on the model relationship being tested, and introduce baselines, variance calibration and diagnostic tests to assess attention's usefulness. Their debate distinguishes finding an alternative from evaluating the explanatory value of the original. Our counterexample concerns an analyst's external selector completion: the common best plan on A2 makes the original-action match nondiscriminating, while A1 separates the completed selectors. It identifies an ambiguity in that validation criterion, rather than testing attention or establishing that the original record is uninformative. This scope follows the broader requirement to evaluate explanations against the system they describe (Jacovi & Goldberg, 2020). Interventions on neural representations address internal causal relations directly (Geiger et al., 2021); our intervention is on the external record and selection rule.

Prediction-based explanation evaluation. Leakage-Adjusted Simulatability (LAS) measures whether an explanation helps an observer predict a model output while controlling for direct output leakage (Hase et al., 2020). REV measures new label-relevant information beyond an input-and-label baseline using condi-

tional V-information (Chen et al., 2023). These approaches motivate assessing what an explanation adds to information already available. Madsen et al. (2024) evaluate self-explanation consistency through counterfactual, attribution and redaction tasks, while Mayne et al. (2026) test added explanation value for predicting a reference model's answers on related inputs. They also use repeated reference-model outputs to assess the prediction ceiling imposed by target variability (their Section 4.3 and Appendix A.7); our single-generation pilot does not estimate that ceiling.

Our prospective pilot applies the added-prediction question to records frozen before precommitted target branches. Both observers already receive the source action; neither sees the target choice before predicting it. The measured quantity is the paired top-1 accuracy difference on later choices across different source and target output protocols. LAS's leakage adjustment and REV's information metric are distinct estimands. The pilot supplies a source-auditable RUU case for this established evaluation question, alongside record-structure and source-action-copy controls.

Connection to scientific discovery. BoxingGym evaluates experiment selection through expected information gain and model discovery through predictions, including whether a scientist's explanation helps another agent predict without further experiments (Gandhi et al., 2025). RUU's present study concerns interpretation of decision records and prediction of model choices. Prediction of withheld environment observations and the utility of selected experiments are the next behavioral targets in our broader research program. Keeping these outcomes distinct connects the measurement lesson here to discovery evaluation without treating action correspondence as discovery success.

8 Interpretation and next research questions

The case study yields a practical sequence for evaluating decision records. First, establish which records support the proposed operation. The transfer panel's zero eligible pairs locate an applicability bottleneck. Second, specify the analyst's completion and test whether the validation cases separate plausible alternatives. The pilot's common best plan shows why a successful original-action match can leave this choice open. Third, report correspondence together with structural opportunities for rule discrimination. The unforced-proposal follow-up demonstrates their coexistence: all sixteen required records match, while fourteen have mathematically forced rule agreement. Finally, evaluate added prediction on new decisions. The prospective pilot supplies that separate measurement and observes no positive net accuracy gain in its assigned cases.

These checks also determine the appropriate unit of evidence. Under the fixed numeric contract, a per-record rule scan exposes the sensitivity in the paired display (Appendix K). Action changes and P/L/U label changes answer different questions, as the two rule-sensitive but label-preserving records illustrate. The proposed audit makes these distinctions inspectable; its effect on downstream diagnostic accuracy remains to be evaluated against independently established diagnoses.

Scope of the evidence. The histories are purposefully selected within two task families. Continuous P/L labels retain the numerical and historical-runtime premises in Appendix F; P certifies a feasible continuation, not a completed task. All completed C proposals already agree with their selectors, so correcting a conflicting proposal has no observed denominator. The older panel's seven analysis errors, including its sole label-changing rule-sensitive record, have no isolated behavioral-effect estimate. Record correspondence and external substitutions address observed outputs; internal causal faithfulness requires a different intervention. In the prospective pilot, same-protocol targets, repeated identical-input targets, and a control that demonstrates use of informative records in the scientific tasks would help distinguish protocol transfer, target variability and observer sensitivity.

Research under uncertainty. Three questions guide the next stage: how a stated assumption relates to the next experiment choice; whether that experiment resolves more uncertainty than an equally resourced valid fixed plan; and whether a revised explanation improves predictions for cases withheld during its construction. Each question calls for its own scored behavioral outcome. The present prospective pilot measures model-choice prediction. Prediction of new world observations and the utility of adaptive experimentation will extend this program from interpreting decision records to evaluating scientific behavior.

9 Reader access and reproducibility

The anonymous supplement supplies the manuscript sources, current-study mechanisms, visible interaction evidence and source-bound readers. Appendix M lists the material and check boundaries for each evidence set. One archive contains both the main and prospective components in a lossless text representation; its reader restores their exact bytes, verifies their manifests and runs the evidence checks. From the extracted supplementary archive, run:

```
python3.12 -I -B -S verify.py
```

The command recomputes the reported source-level results, checks the paired prediction outcomes and selector-discrimination example, and runs the existing numerical replays. No model account is needed. The README also provides permanent extraction and corruption-test commands. Source recalculation and present-host numerical replay establish reproducibility of the supplied evidence; historical/provider attestation and independent laboratory replication remain separate verification levels.

Responsible interpretation. The experiments use simulated environments and model outputs. The checks evaluate external decision records; assessing internal reasoning, intentions or deployment safety requires evidence for those distinct claims. The disclosed mechanisms support inspection of this fixed study. Future benchmark mechanisms have a separate disclosure policy.

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A Disclosed task, mechanism and delivered information

Represent the two-dimensional state and action by complex numbers z_t and u_t. A step is indexed by its pre-action state: u_0 produces z_1. The initial conditions are $z_0 = 1$, $\phi_0 = 4.147589101011308$ and $u_{-1} = 0$. The evaluated world obeys

$$d(z) = (1.06 + 0.015|z|^2)z + (0.12 - 0.08i), \tag{1}$$

$$z_{t+1} = d(z_t) + e^{i\phi_t}u_t, \qquad |u_t| \le 0.6, \tag{2}$$

$$\phi_{t+1} = \phi_t + \omega + \kappa \,\mathrm{Re}(u_{t-1})\,\mathrm{Im}(u_{t-1}), \qquad \omega = 0.1. \tag{3}$$

The lag in the phase update is part of the mechanism: the update uses u_{t-1}, not the action just chosen. The original family allowed $\kappa \in \{0.3, 3, 6, 12, 18\}$. Only the two settings below occur in the fixed-history pilot. They are parameter settings, not a validated difficulty ordering.

Paper alias	Archival case identifier	Snapshot after step	κ
A	case-01	4	12
B	boundary-01	11	12
C	case-03	4	18
D	boundary-02	9	18

An episode ends on escape ($|z_t| > 4$) or after 32 actions. Success requires completion and $|z_t| \le 0.05 + 10^{-10}$ at *all* $t \in \{28, 29, 30, 31, 32\}$. Merely avoiding escape is insufficient. Action validation rejects nonfinite coordinates or norms above $0.6 + 10^{-9}$; a smaller tolerance overshoot is normalized to 0.6. State observations are rounded to twelve decimal places, whereas the simulator retains binary floating-point state. Stored actions retain their applied precision. The exact numeric history of every selected case is included in `legacy-replay/data/cases.json`; it is not regenerated from a caption or rounded manuscript table.

The participant receives the objective, horizon, action limit, escape threshold, full permitted action–state history and remaining analysis allowance. It does not receive these equations, ϕ_t, ω, κ, the evaluator's continuation or its loss diagnosis. The opaque identifiers in delivered inputs do not disclose the boundary/comparison labels. The frozen selector and schemas are inspectable in the reader artifact; the full provider logs are not redistributed. Disclosure to reviewers does not alter the historical participant information boundary.

B Witness and loss-bound justification

Positive witness. For every pre-action snapshot and every resulting state, one fixed privileged controller attempts a continuation. At each step it chooses

$$u = \mathrm{clip}_{0.6}\left[-d(z)e^{-i\phi}\right], \qquad \mathrm{clip}_r(v) = \begin{cases} v, & \|v\| \le r, \\ \dfrac{rv}{\|v\|}, & \|v\| > r. \end{cases} \tag{4}$$

Here $\|v\|$ is the Euclidean norm, equal to the modulus for a complex scalar. The original transition, action validation, escape check and scheduled target test score the entire continuation. A successful admissible continuation establishes existence for the implemented task. A failed continuation does not establish impossibility. This controller knows the hidden phase and mechanism; its success is not evidence of blind action identification or participant learning. Every pilot pre-action state passed this positive baseline.

Sufficient negative bound for the ideal recurrence. Write $x = (\mathrm{Re}\, z, \mathrm{Im}\, z)$ and $c = (0.12, -0.08)$. Fix a vector q with $\|q\|_2 \le 1$. Suppose $q^T x \ge p > 0$. Since $\|x\|_2 \ge q^T x \ge p$, and the scalar multiplier $1.06 + 0.015\|x\|_2^2$ is positive, the next-state projection satisfies

$$q^T x' = (1.06 + 0.015\|x\|_2^2)q^T x + q^T c + q^T R(\phi)u \tag{5}$$

$$\ge (1.06 + 0.015p^2)p + q^T c - 0.6. \tag{6}$$

The last term follows from Cauchy–Schwarz and the admissible action radius. The function $g(p) = (1.06 + 0.015p^2)p$ is increasing on the positive domain. Consequently, iterating this lower bound while it remains positive gives a bound for *every* allowed future control sequence, irrespective of the future phases. A bound exceeding 4 before the horizon implies unavoidable escape; a bound exceeding the target tolerance at a required target step excludes successful completion. A target violation already present in the recorded prefix also suffices, because future actions cannot change a past required state.

Implementation and numerical scope. The evaluator chooses q by truncating each coordinate of $x/\|x\|$ toward zero to four decimal places, and checks its norm. It initializes $p_0 = q^T x$ from the stored full-precision state. The implemented recurrence is

$$p_{j+1} = \left\lfloor (1.06 + 0.015p_j^2)p_j + q^T c - 0.6 - 10^{-9} \right\rfloor_{10^{-12}}, \tag{7}$$

where $\lfloor \cdot \rfloor_{10^{-12}}$ rounds downward to that grid. Coefficients and initial state are converted from their actual binary-float values to exact rational numbers. Loss is reported only for a positive induction sequence crossing 4.000001 within the remaining horizon, or crossing $0.05+10^{-10}$ at a required target step. A separate 110-digit decimal implementation checks the direction, every bound, the positivity domain, threshold and horizon. The code also checks that the transition definition and task constants match the bound's assumptions.

Archived arithmetic audit. A subsequent record-only audit covers the pilot, distinct component grades and the continuous subset of the older 48-trial transfer panel. All 50 distinct stored directions, including directions for which no loss was certified, have exact squared norm at most one; the minimum value of $1 - \|q\|_2^2$ is 5.75×10^{-6}. There are ten distinct certified projection records (eleven occurrences). Rational recomputation reproduces the archived binary64 bounds exactly. Recomputing these recurrences with per-step subtractions 10^{-9}, 2×10^{-9} and 10^{-8} preserves every certificate kind and first certification step. Even at the enlarged slacks, the smallest final lower-bound gap above the applicable runtime threshold exceeds 0.0371. No simulator transition or model request is performed by this audit, and no historical certificate is rewritten.

The action validator rejects computed norms above $0.6 + 10^{-9}$ and rescales a smaller computed overshoot by $a\,0.6/|a|$. It therefore does not simply apply every action up to the rejection tolerance unchanged. For the 1,201 distinct applied actions in the inspected continuous trajectory records, exact squared-norm checks give a maximum norm of approximately 0.60000000000000002694. This finite record check does not bound all possible future actions or transcendental-function errors.

Runtime-specific conditional budget. A separate local audit inspects the pinned interpreter and the currently loaded ARM64 math image. Its compiled complex multiplication uses fused multiply-add instructions; its complex exponential calls `exp` and `__sincos_stret`. The norm uses platform `hypot`, and the square still uses platform `pow`. Thus a source-level list of separate sine/cosine calls does not fully describe this executable.

Appendix F derives a norm bound for the captured `hypot` instruction sequence under ordinary IEEE binary64 round-to-nearest semantics with gradual underflow. It then reduces the remaining library accuracy requirements to an absolute square error ϵ_{pow} and a norm excess δ_{rot} for the computed complex rotation. Accuracy relative to the true angle is not needed for this projection bound. Taking each of these two bounds as 2^{-30} yields total one-step error below 6.288×10^{-10}, which fits the archived 10^{-9} subtraction. The local checker verifies the scalar inequalities with exact rational arithmetic.

This is a sufficient conditional bound, not an attestation that the two library bounds hold uniformly. Finite current-process probes satisfy them but do not prove them. In addition, the two inspected historical runtime inventories and activation records do not pin the loaded math-image identity or floating-point control register. Today's image capture cannot retroactively fill that gap. The ten archived projection certificates therefore retain their stated conditional numerical status. The route family's exact integer transitions and finite-horizon dynamic program do not use this projection premise.

The classifier returns preserved only with a successful witness, lost only with a sufficient negative certificate, and unresolved otherwise. Simultaneous positive and negative evidence is treated as an error. This is an

incomplete classifier, not a complete viability kernel or a map of a continuous safe cone. A numerical failure in candidate optimization is a different unresolved category and is not fed to this classifier as physical loss.

C Computational interventions and reproducible selectors

The candidate coefficients and original analysis outputs are fixed data. No candidate is refitted after a component is crossed. Let $f_m + B_m u$ denote candidate m's predicted next state at the selected snapshot. Candidate construction, feature families, phase extrapolation and the exact numeric coefficients are inspectable in `legacy-replay/data/component_inputs.json`. The replay implements these fixed operations; it does not execute the archived participant programs.

One-step worst-case rule. For G equally spaced angles and radius r, set $u_j = r(\cos(2\pi j/G), \sin(2\pi j/G))$ and choose the first grid index minimizing $\max_m \|f_m + B_m u_j\|_2$. Snapshot B uses $G = 720$ and $r = 0.6$; D uses $G = 1440$ and $r = 0.599999999$. The four finer-grid checks use $G = 2880$. This is optimization on a finite circle, not a guarantee of global optimization over the full action disk or of task completion.

Past-error weighted rule. Each fitted parent has weight $w_m = 1/(10^{-4} + e_m^2)$, where e_m is its recorded leave-one-out residual RMS. In the expanded D component, six scenarios inherit equal shares $w_m/6$ of each parent's weight. With $v = \sum_m w_m B_m^T f_m$, the selected action is $u = -0.6v/\|v\|$. The matrices here are scalar gains times rotations, so the quadratic term in $\sum_m w_m \|f_m + B_m u\|^2$ is constant on the radius-0.6 circle. Thus this direction minimizes that fixed-radius surrogate for these matrices. It is not a general theorem for arbitrary linear-control matrices, a calibrated posterior, or a worst-case task-safety guarantee. D's second representation comprises 11 parent fits expanded by two phase modes and three phase offsets $\{-0.12, 0, 0.12\}$; the complete construction is in the code and numeric inputs.

Multi-step plans and the researcher completion. For each of B's candidates, the individual planner optimizes the remaining $N = 32 - n$ controls with projection to the action disk. Its surrogate objective is

$$J_m = \sum_{i=1}^{N} \left[w_{n+i} \|\hat{x}_{n+i}\|^2 + \lambda \big(\max\{0, \|\hat{x}_{n+i}\|^2 - 14.44\} \big)^2 + 10^{-5} \|u_{n+i-1}\|^2 \right], \tag{8}$$

where $w_t = 1$ for $t \geq 28$ and 10^{-4} otherwise, and $\lambda = 10$. The deterministic initialization, projected accelerated-gradient updates, backtracking limits and restart schedule are included in `components.py`; individual plans use 1,000 iterations. The declared selector R2* chooses the individual plan with the lowest computed cost and takes its first action. This final selection rule was completed by the researchers. Agreement with the original choice is a reconstruction check, not proof that it was the model's latent policy. At least one crossed action has magnitude approximately 0.5755, so this contrast does not isolate direction or horizon while holding magnitude fixed.

The two auxiliary joint plans use a shared first action, separate candidate-specific later actions, the mean of the candidate costs, and 1,400 iterations. One is numerically unresolved. The one-at-a-time D deletions remove each of 66 scenarios or each of 11 six-scenario parent groups; all other settings stay fixed and no refitting occurs. No-rescue results cover only these interventions, not joint deletion, changed fitting or arbitrary alternative uncertainty models.

The 91 planned cells comprise 8 primary crossings, 4 finer-grid checks, 2 joint-plan auxiliaries and 77 omissions. Ninety were computed and one auxiliary remained numerically unresolved. Repeated actions are deduplicated before privileged grading, giving 13 distinct grades. There is no additional model sample in this count. The full records, rather than the rounded values in figures, define action identity.

C.1 Five retrospective completions at pilot context B

This separate sensitivity analysis holds the eight individual plans from B's native-loss pool A1 and four from its native-preservation pool A2 fixed. It changes only which plan supplies the first action. For a plan's predicted states $\hat{x}_t$, the five minimized criteria are: the total penalized cost J_m above; $\max_{28 \leq t \leq 32} \|\hat{x}_t\|$;

$\frac{1}{5}\sum_{t=28}^{32}\|\hat{x}_t\|^2$; $\max_{n\le t\le 32}\|\hat{x}_t\|$, including the common starting state; and $\|\hat{x}_{32}\|$. All ties use the lowest existing plan index. The last criterion covers only the final target step; it is not the full task-success condition. The finite selector list was fixed before its additional grading, with the original outcomes already known.

Table 6 reports these five selectors in the main text.

Thus native-action reproduction alone does not identify the crossed completion. The reversal also changes the chosen plan and action magnitude; it does not isolate a horizon-only or direction-only effect. The saved selector indices, exact actions, metrics and labels are in `artifact/source/pilot-sensitivity.private.json`. The original action lock and full twelve grade records, including the fixed-controller continuation trajectories and projection certificate, are supplied alongside it as `pilot-sensitivity-actions-lock.json` and `pilot-sensitivity-grades.json`. Their original hashes are checked before replay. The standard reader uses recomputed plan actions as grader inputs, binds the case, history, world, action and certificate state, and compares the complete replayed grade with the archive. Discrete fields and identities must match exactly; floating trace values and plan metrics use $|a-b|\le 10^{-8}(1+|b|)$, while native-action compatibility uses 10^{-6}. Each projection certificate also passes the separate 110-digit arithmetic check; this does not prove the uniform library bounds or historical runtime premises in Appendix F. The reader reports its twelve verification grade calls separately from the original collection counters. This finite sensitivity result does not cover new optimizers, mixtures or arbitrary alternative policies.

D Protocol, record boundaries and cost accounting

The fixed trial order was A1, D1, C1, B1, C2, B2, A2, D2, where the suffix denotes repeat. Each trial begins in a fresh local context with an opaque snapshot identifier; both repeats of a case receive exactly the same initial payload bytes. Each subsequent payload contains the same physical history and that trial's own submitted analysis code and returned results. Therefore the third or fourth requests are not equal-input repetitions of the first request.

The wire schema permits one analysis request or one final two-coordinate action. Analysis code must define `act(history)` and can use the allowed mathematical facilities; no network, filesystem, process creation or hidden simulator interface is provided. The protocol and policy source state the exact permitted syntax and names. Local analysis has 30-second CPU/wall limits, a sampled 1 GiB RSS stop and a 100,000-character output limit. Sampled RSS is not a hard allocation ceiling. Each trial permits at most three analysis requests and one final action, for at most four model requests in total. Trial wall time is 3,780 seconds; request time is bounded by the smaller of 900 seconds and remaining trial time minus a 120-second record/grading margin. No retry, neutral request, replacement or outcome-dependent stopping is permitted.

The model identifier and effort are configuration records, not attested server weights. Local receipts record one agent message per call, no unexpected tool events and cleanup of initially empty call directories. These checks do not establish independent sampling seeds or provider-side isolation. Full provider transcripts remain in the source archive; this bundle distributes selected numerical inputs and final record projections. Hidden chain-of-thought is not observed or claimed as evidence.

The recorded 588,051 input and 129,677 output tokens total 717,728. The 92,160 cached-input tokens are already included in input. Reasoning-output tokens are likewise a subset of output. There were 28 receipted requests and zero unreceipted requests in this pilot. Provider receipts are the source of these counts; analyst use and money are unmeasured. The historical technical null's unresolved reservation is not a zero-use observation and is not included in the pilot total.

E Prospective tasks, reference histories and adapter

Exact configurations and twelve packets are in `legacy-replay/data/transfer/`; the numeric replay inputs under `legacy-replay/data/transfer/` are checked against these records.

Continuous reference histories. The recurrence and task target are those in Appendix A, with three new initial phase/frequency configurations and couplings 3, 6 and 12. Phases and frequencies derive deterministically from SHA-256 of the literal seed `transfer-validation-20261025-v1` and instance ordinal. A prescribed privileged ten-step path tracks waypoints with radii $1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.75, 1.9, 2.05, 2.2$ and deterministic changing angles. Snapshots are after steps 6 and 10. Pre-collection checks require legal histories and a positive continuation, without resampling. This selection does not establish calibrated boundary distances or natural participant behavior.

Route–resource dynamics and certificates. The state is node $n \in \{0, \dots, 18\}$ and remaining integer fuel f. Action $j \in \{0, 1, 2\}$ is encoded as $[j, 0]$ and advances to $n' = \min(18, n+j+1)$. The hidden stationary positive edge cost is

$$c(n, j) = b_j + ((n + o_j) \bmod p_j), \qquad f' = f - c(n, j). \tag{9}$$

Success means reaching node 18 within twelve actions without negative fuel. The three fixed (b, o, p) configurations and initial fuel values are in `CASES.private.json`. Unlike continuous control, every legal route action is in the menu. The participant knows the graph, objective, horizon and observed costs through its action–state history, but not the edge-cost formula or private coefficients.

Let $V_h(n)$ be the minimum fuel needed to reach the goal in at most h further actions. Set $V_0(18) = 0$, $V_0(n < 18) = +\infty$, and

$$V_h(18) = 0, \qquad V_h(n) = \min_{j \in \{0,1,2\}} \{c(n, j) + V_{h-1}(\min(18, n + j + 1))\}. \tag{10}$$

The code represents infinity by `None`. With positive costs, $f \geq V_h(n)$ provides a minimum-cost admissible continuation; $f < V_h(n)$, negative fuel or an unreachable goal certifies loss. Every value-table cell is checked by a separately implemented recursive recurrence. Positive witnesses are replayed. Initial fuel is the exact minimum completion cost plus one; the lexicographically first minimum-cost path supplies snapshots after steps 2 and 4. Baseline feasibility does not prove that the hidden cost law is identifiable from the observed short prefix.

Exact optional table contract. The diagnostic continuous menu has twelve equally spaced directions at radius 0.6, rounded to twelve decimals, plus zero. For a returned table with losses L_{mj} and weights w_m, the weighted rule chooses the first menu index minimizing $\sum_m w_m L_{mj} / \sum_m w_m$; the worst-case rule minimizes $\max_m L_{mj}$ with the same tie break. Weights belong to the candidate pool. The adapter accepts only the exact keys `actions`, `candidate_losses`, `weights`, and `rule`, the exact delivered menu, 1–128 candidate rows and finite entries of absolute magnitude at most 10^{100}. Weights are finite, nonnegative and bounded by 10^{100}, with positive total. Final-action reproduction uses maximum coordinate error $\leq 10^{-8}$. The last successful analysis is selected before examining outcome category. A successful earlier table cannot rescue a missing or unfaithful final table.

The replay changes only imports/data paths, preserves frozen numeric functions, and omits dispatch code. Original function text and source/export digests are included. It reads bound archived analysis values; participant programs remain unexecuted text. It reproduces the eligibility test without inferring or regenerating tables.

F Narrow runtime proof audit

This appendix states a sufficient numerical contract for the continuous projection argument and documents which parts were established locally. The real-arithmetic argument, the archived recurrence computation, the current machine implementation and the historical execution environment are distinct evidence layers. None of the checks below adds a live agent trial or reruns a physical episode.

Identity and trust assumptions. The inspected executable matches the experimental Python digest. It is CPython 3.12.14 on ARM64. The current math image is `libsystem_m.dylib`; its UUID, text/constants hashes and relevant function bytes were recorded. The current thread reports a zero floating-point control

register: round to nearest, with no flush-to-zero or alternative handling enabled. We assume the ordinary IEEE binary64 semantics of the decoded add, multiply, divide, fused multiply-add and square-root instructions. The source-to-instruction interpretation is a manual derivation with exact constant checks, not a proof-assistant-certified binary analysis. The historical inventories pin Python but do not attest the loaded math image or control register; equivalence to the historical environment remains an explicit assumption.

Bound for the captured norm implementation. Let $u = 2^{-53}$ and $d = 2^{-1075}$, the unit roundoff and half the smallest subnormal. Here u denotes roundoff, not a physical action. For finite binary64 coordinates, put $a = \max(|x_1|, |x_2|)$ and $b = \min(|x_1|, |x_2|)$, and let $r = \sqrt{a^2 + b^2}$ in exact arithmetic. The captured function returns a when $b = 0$. Otherwise it drops b only if the stored exponent difference exceeds 27. This implies $b/a < 2^{-27}$, including a subnormal b, so dropping it reduces the exact norm by less than $r2^{-55} = ru/4$.

For nonzero retained inputs, the decoded branches use the following scales, where e_a is the biased exponent field of a:

Exponent field	Input scale	Undo scale
$e_a < 539$	2^{768}	2^{-768}
$539 \le e_a < 1534$	1	1
$e_a \ge 1534$	2^{-768}	2^{768}

The retained inputs scale exactly. In the small-input branch even the smallest subnormal becomes 2^{-306}. In the large-input branch a nonzero retained b has exponent difference at most 27 and remains normal after scaling. In all branches the squared larger input and the sum are normal and cannot overflow. The core computes $t = \mathrm{RN}(a_s^2)$, $v = \mathrm{RN}(b_s^2 + t)$ by a fused operation, and $w = \mathrm{RN}(\sqrt{v})$, before undoing the scale. Thus w lies within factors $(1-u)^2$ and $(1+u)^2$ of the scaled norm. Undoing a power-of-two scale is exact for a finite normal output; subnormal rounding contributes at most d. Overflowed outputs are not finite accepted norms. Combining the discarded-input allowance gives the sufficient bound, for a finite returned norm h,

$$r(1-3u) - d \;\le\; h \;\le\; r(1+3u) + d. \tag{11}$$

The rational checker verifies $(1-u/4)(1-u)^2 \ge 1-3u$ and $(1+u)^2 \le 1+3u$. In particular, $h \le 4$ implies $r < 5$ and $|h - r| < H = 16u$. This is an analytic bound for the captured sequence under the stated instruction semantics, not an accuracy inference from test samples.

Action normalization. Let L be the exact binary64 value of 0.6. An accepted but unscaled action satisfies $h \le L$, hence $r \le (L+d)/(1-3u) < L + 8u$. When $h > L$, the validator rescales by aL/h. Here $r > 1/2$ and $h > 1/2$. The compiled multiplication by a real scalar and division by a positive real denominator each round each component once; the zero imaginary terms do not introduce a separate nonzero error. Using $\sqrt{2} < 3/2$ gives

$$\|a_{\mathrm{applied}}\|_2 \le \frac{L(1+u)^2}{1-3u-2d} + 3d(1+u) + \tfrac{3}{2}d < L + 8u. \tag{12}$$

This covers accepted rescaling, including subnormal allowances, rather than treating the rejection tolerance as an applied-action allowance.

Two remaining library accuracy contracts. Write $\widehat{s} = \mathrm{pow}(h, 2)$ and $C(\phi) = \mathrm{cmath.exp}(i\phi)$ for the computed values. The needed contracts are

$$|\widehat{s} - h^2| \le \epsilon_{\mathrm{pow}}, \qquad \|C(\phi)\|_2 \le 1 + \delta_{\mathrm{rot}}. \tag{13}$$

They must hold uniformly over relevant norm inputs and permitted finite phases. The second includes the wrapper's `exp` and combined `__sincos_stret` computation. No bound on angular error is needed: every such control vector has norm at most $(1+\delta_{\mathrm{rot}})(L+8u)$, irrespective of its direction. Inspection of call targets establishes the operation path but does not prove either accuracy contract.

Composed one-step bound. For a state before computed escape, $r < 5$. Let α, β, c, L denote the exact binary64 task constants, so the reference drift is $(\alpha + \beta r^2)x + c$. Set $B = 32u + d$. Under the sufficient choices $\epsilon_{\text{pow}} = \delta_{\text{rot}} = 2^{-30}$, the relevant basic-operation intermediates have magnitude below 32; the square is handled separately by its library contract. Each basic rounding error is therefore at most B. The squared-radius error is bounded by $10H + H^2 + \epsilon_{\text{pow}}$. Two basic operations form the drift multiplier. Scaling the state and adding the bias give vector error at most $5\beta(10H + H^2 + \epsilon_{\text{pow}}) + 13B$. The compiled complex product uses one ordinary product and one fused operation per component; its error contributes less than $3B$. Final vector addition contributes less than $3B/2$.

Consequently the sufficient loss allowance beyond the ideal $-L$ term is

$$E = 5\beta(10H + H^2 + \epsilon_{\text{pow}}) + 18B + 8u + \delta_{\text{rot}}(L + 8u). \tag{14}$$

The exact rational evaluation is approximately $6.2870891 \times 10^{-10} < 10^{-9}$. For $\|q\|_2 \le 1$ and a positive projection lower bound p, this yields

$$q^\top x' \ge (\alpha + \beta p^2)p + q^\top c - L - E.$$

The archived subtraction and downward floor therefore remain sufficient *if* the stated instruction, library and historical-equivalence premises hold. The prior audit's minimum terminal gap exceeds 0.0371; it also clears the norm-test error. The same ten unique archived projection certificates are covered conditionally, with no relabeling.

Checks and open obligations. The archived local audit checked exact inequality constants, captured function identity and historical inventory coverage. The reader bundle rechecks only the exact constants and exponent ranges with `artifact/numeric_constants.py`; it does not recapture or attest the historical runtime. Fifty-two norm inputs, including exponent/underflow edges, twenty square inputs and twenty-six phases were probed in the archived audit interpreter. Every finite tested output satisfies the sufficient bounds; an overflowed norm is recorded separately. These are scalar diagnostics, not uniform accuracy proofs. Negative controls reject altered function bytes and oversized error budgets, and retain an open status when a library premise or historical binding is missing. They test the checker, not the missing premises themselves.

The remaining obligations are explicit: prove both bounds in Eq. (13) for the relevant library implementation, and justify its equivalence and floating-point settings in the historical runs. Current image hashes cannot supply absent historical attestation. Accordingly, this addendum narrows the numerical assumptions and supports their composition; it does not turn the archived results into unconditional machine-verified claims. Exact route dynamic programming and the observed trial counts are unchanged.

G The 32-trial common-menu protocol and panel usage

Selected histories and menus. Table 15 identifies the four purposeful histories using the earlier transfer panel's source IDs. The two continuous histories are at physical step 10; the two route histories are at step 2. Each appears with both profiles, both repeats and both conditions. These are selected reference histories, not a random sample of histories or eight newly sampled worlds.

Delivered ID	Source ID	Family	Instance	Cutoff	Menu size
probe-01	snapshot-03	Continuous	1	10	13
probe-02	snapshot-07	Continuous	2	10	13
probe-03	snapshot-02	Route	1	2	3
probe-04	snapshot-06	Route	2	2	3

Table 15: Fixed histories for the 32-trial panel. Source histories and menu identities are hash-bound in the archived allocation.

Both conditions receive identical history and task/menu bytes for a given probe; the condition instruction differs. The continuous menu is the twelve radius-0.6 directions at angles $2\pi j/12$, $j = 0, \ldots, 11$, rounded

to twelve decimals, followed by zero. The route menu is $[(0,0),(1,0),(2,0)]$ in that order. Only the indexed menu action is executed. The original physical clock, remaining horizon and task objective are unchanged. The participant receives visible history, public task/menu and its own analysis outputs, without hidden dynamics or future observations. Trial order was fixed before the run and is distributed in `artifact/data/crossing.json`.

Common output scaffold and condition-specific execution. The shared structured-output scaffold wraps a `decision` object that can request analysis or return an action ID, a record, or both. It does not itself enforce condition-specific fields, record identity, dimensions, loss range or action agreement; the local contract does. A record has exactly six fields: `schema`, `case_id`, `menu_sha256`, `loss_definition_id`, `candidate_losses` and `rule`. It contains 1–128 rows, one numeric loss in $[0,1]$ per menu action, and no weights. Each row uses the same declared meaning: an uncalibrated forecast of failing the original full task after that next action and the best continuation the participant can plan. The two rules minimize the uniform row mean or row maximum. Decimal numeric tokens are evaluated as exact rationals; ties select the lowest menu index. Single and duplicate rows are legal, and duplicates retain multiplicity. Rows are not verified world models.

In the optional condition, a legal `action_id` is required and an accompanying record is optional. Record failure or disagreement is logged without replacing that action. In the mandatory condition, only the record is allowed; its declared rule deterministically selects the executed action. An invalid record produces no action and no retry. Thus mandatory agreement is by construction, while optional agreement is an observational check. This six-field contract differs from the 48-trial panel's weighted, four-field last-analysis adapter in Appendix E; their eligibility counts are not measurements of the same interface.

Budgets and recorded use. For the 32-trial panel, each fresh trial permits up to three optional local analyses and one final decision: at most four model requests, 3,780 seconds per trial and 900 seconds per request. Analysis has a 30-second CPU/wall limit, sampled 1 GiB RSS stop and 100,000-character JSON output cap. It yields no new physical observation. No retry, repair prompt, neutral request or outcome-dependent extension is allowed; the panel ceiling is 128 model requests. The completed panel used 48 analysis requests and 32 final decisions. Its single selected action per trial was graded; crossed labels reuse the bound menu census, not additional live observations.

Panel	Requests	Input	Output	Cached input	Unknown calls
48-trial transfer	120	2,325,280	484,134	372,352	0
32-trial common menu	80	1,733,978	422,952	263,168	0

Table 16: Separate participant-model usage ledgers. Cached input is already included in input, and reasoning tokens, when reported, are part of output. Neither category is added a second time. Pilot usage is in Appendix D.

Table 16 reports archived receipt-derived counts, not analyst usage or provider charges; these latter quantities were not measured. Synthetic transport and privileged local computation are outside the table. The 32-trial plan reserves 1,050,000 input plus 128,000 output tokens for each request lacking usage, separately from measured totals. Its final unknown-call count is zero, so no such reserve remains in that panel's final accounting; this policy is not a provider billing cap. The source-bound scalar projection is `artifact/data/protocol-summary.json`. It provides provenance and protocol identities without redistributing the full provider logs.

Exact requests and executable reader path. `artifact/protocol/` contains both condition instructions, the shared wire schema, the four initial packets and all 80 request payloads, including each trial's accumulated analysis records. Its checker renders every full prompt and verifies the archived prompt and payload hashes. For example,

```
python3.12 -I -B -S artifact/protocol/check.py --show-call 1
```

prints the first request. The historical collection entry point is supplied as inspection-only text; provider transport credentials and full provider responses are not redistributed. Configured profile names are not independent attestation of server weights.

`artifact/menu_replay.py` binds each public packet to its transfer history and menu, then recomputes the 26 continuous and six route menu labels. The original record comparison and this numerical replay are separate checks. Continuous loss labels retain the open uniform-library and historical-runtime premises in Appendix F; present-host agreement does not discharge them.

H Study inventory and historical boundaries

The eight-trial pilot, 48-trial prospective transfer panel, 32-trial common-menu panel, separate 48-assignment follow-up and new 16-target prospective choice-prediction pilot are five evidence sets in the main paper. The two 48-slot studies are distinct. Their counts are never pooled. The follow-up has 46 completed trials and two technical losses, not 48 completed scientific trials. The two 1,600-draw development samples and the 5,000-draw main acquisition study appear only in the companion; the earlier prediction criteria failed and are retained there. The acquisition result is not an additional live LLM or crossing experiment.

Older archive labels describe different work: *H4* denotes an unfinished collection whose 84 remaining assignments stayed paused and whose crossing status is unassessed due to missing evidence. The *historical technical null* denotes an earlier infrastructure-limited record, not a successful zero-use scientific trial; its unresolved usage reservation stays outside the pilot accounting. These records supply no extra evaluable trials here. The archive's v1.8/v1.9 labels identify preserved predecessor research versions, not additional studies or independent replications. Current paper revision numbers track presentation and reader-access changes. This revision additionally incorporates the prospective choice-prediction pilot, with its original interrupted segment and separately completed last target block. The earlier follow-up, predecessor scientific inputs and denominators, and the primary acquisition procedure remain unchanged. The world-prediction research design is unrun and contributes no result to this manuscript.

I Analysis-interface errors and consumed slots

The historical shared instruction describes `history.states` and `history.actions`; the evaluator actually supplies a Python dictionary. Five analysis calls failed on `history.states`. Two other calls failed because `next` is absent from the restricted built-ins; the historical instruction refers to the existing interface without listing those built-ins. These are observed interface-use errors, not missing final decisions. The seven affected trials are listed below. Each made three analysis attempts: one failed and two succeeded. Errors consumed an attempt and were visible to the next request. They did not trigger replacement trials or extra slots.

History	Profile	Condition	Repeat	Error
probe-01	gpt-6-astra	Mandatory	1	Dictionary attribute
probe-01	gpt-6-astra	Optional	2	Dictionary attribute
probe-01	gpt-6.1-sol	Mandatory	2	Dictionary attribute
probe-01	gpt-6.1-sol	Optional	1	Unavailable `next`
probe-01	gpt-6.1-sol	Optional	2	Unavailable `next`
probe-02	gpt-6-astra	Optional	1	Dictionary attribute
probe-02	gpt-6.1-sol	Optional	2	Dictionary attribute

Table 17: All seven affected common-menu trials. All are continuous-task trials; both profiles requested max reasoning effort. The first row is the sole rule-sensitive mandatory record. These are descriptive associations, not estimates of the errors' effects on actions or records.

The 32 common-menu trials produced 48 analyses: 41 successful and seven erroneous. In the separate transfer panel, all 24 route trials made zero analysis attempts, while the 24 continuous trials made 72 attempts: 67 successful and five erroneous. These denominators are not pooled. The frozen adapter still supports only one transfer record and no complete pair.

`artifact/interface_audit.py` reconstructs all 32 trial rows from the last cumulative analysis list in each trial's archived requests, checks every earlier list as a prefix, and reconciles attempt counts with the final trial result and remaining-slot field. It also accounts for the 48 transfer trials. No archived submitted program is executed. This procedure avoids counting the same error again each time it appears in a later prompt.

The reconstructed tables and their crossing identities remain checkable. However, the observed scarcity and row structure occurred under this imperfect interface. A corrected description could change analysis use, record structure or actions; no such behavioral effect has been measured. Local checks of dictionary access and built-in availability cannot answer that causal question. A prospective comparison must preserve old records and compare the two interface descriptions under a separately specified allocation and budget. Mandatory reporting and external execution would still be a combined intervention in either interface arm.

Pilot sensitivity reader boundary. `artifact/pilot_sensitivity.py` regenerates all twelve single-candidate plans and their five selection metrics from the disclosed component inputs, then reconstructs the five selected indices and native-action compatibility. It now re-evaluates all twelve first actions and compares the full archived witness/certificate records as described in Appendix C.1. This extends the reader through the conditional task labels. The numerical and historical-runtime premises of the grade evidence remain open.

J Follow-up contract, accounting and verification boundary

The follow-up profiles are gpt-6-astra/max/Fast and gpt-6.1-sol/max/Fast. Each has 24 original assignments, eight per arm and eight matched triplets. The same four historical contexts are used, with two repeats per profile; neither repeats nor individual requests are new independent task samples. A and B execute a legal proposal. A may omit its record; an invalid optional record does not invalidate an otherwise legal proposal. B and C require a valid record and legal proposal; a protocol-invalid final produces no action. C executes the declared record selector and retains the proposal. The common schema checks case/menu identity, dimensions, finite values and legal actions. Single rows, duplicates and same-action rules are allowed. The shared analysis interface uses dictionary access and states its available operations, addressing the earlier interface mismatch; this is not a randomized old/new interface comparison, so its causal effect is unmeasured.

Each trial permits at most three optional analyses and one final decision, four model requests, 900 seconds per request and 3,780 seconds per trial. Errors consume a slot; model responses are not repaired and failed assignments are not replaced. The closed results record 36 analyses for Astra and 36 for Sol, with zero recorded analysis errors in each. These counts do not turn the historical interface comparison into a controlled experiment. Decisions were bound before privileged label lookup. Follow-up labels reuse the same identity-bound finite menu census, retaining the continuous numerical and historical-runtime premises of the earlier paper. No new physics or live crossed model decision was generated for this analysis.

Three execution segments and missingness. The original 48-position allocation was retained across three separately authorized execution segments. Positions 1–12 completed; 13 and 14 ended with HTTP 429 technical losses in the first and second segments. The third segment completed positions 15–48 without repeating either failed position. Both technical losses are Astra, in A and C of the same matched triplet. Thus Astra has 22 completed, two technical and zero protocol losses; Sol has 24 completed and zero technical or protocol losses. All 48 assignments started and have result records; none remains active or unstarted. Completion of the 34-position last segment does not mean all 48 assignments scientifically completed. Missing pairs are excluded only from observed-pair descriptions, with their denominators visible; no missing-at-random claim is made. The final segment's 88 request receipts contain no terminal 429 diagnostic. That observation does not identify the earlier limiter or establish a permanent resolution; credit exhaustion is not inferred.

Using the frozen tariffs, measured credit equivalents are 1,336.90190 for Astra and 264.37032 for Sol. Astra's two unknown requests retain a separate 1,690 credit-equivalent reserve, based on 1,050,000 input plus 128,000 output tokens per request. These are local accounting quantities, not a provider invoice or provider token ceiling. The original 7,000 local admission bound was not increased. Measured usage and unknown reserves

Profile	Requests	Known/unknown	Input	Cached input	Output
Astra	60	58/2	1,428,745	311,168	305,022
Sol	60	60/0	1,418,971	263,424	294,997

Table 18: Follow-up accounting only. Token counts cover known usage. Cached input is included in input; reasoning output is included in output. Neither subset is added again. Requests are not independent trials.

remain separate; cache splits are measured for known receipts, while any cold-price treatment of an unknown split would be an assumption, not measured zero cache.

Schema denominators. The retained field `partial_triplets` counts planned blocks missing an A, B or C assignment; its zero value does not assert scientific completion. The reader separately reports fully completed triplets (Astra 7, Sol 8), scientifically incomplete triplets (Astra 1, Sol 0), and triplets containing technical losses (Astra 1, Sol 0). The retained field `cache_split_unknown_requests` counts unknown cache splits only among requests with known token usage: its assessed denominators are 58 for Astra and 60 for Sol. Astra's two usage-unknown requests are excluded from that count and retain their full separate reserves. Thus zero unknown cache splits is neither zero cached input nor known usage for every request. The machine-readable `artifact/followup/SCHEMA.json` defines these populations and exclusions explicitly.

Reader inputs and remaining scope. The package contains 46 exact minimal finals, including 31 original numeric tables and declared rules, plus the common/A/B/C instructions and wire schema. The reader checks trial, profile, block, case and menu identities; reparses decimal tokens as exact rationals; recomputes rule selections and row diagnostics; and reconciles decisions, commit hashes and aggregates. It checks A/B against the proposals and C against the record selector. All 46 completed labels are re-read from the bound `artifact/data/crossing.json`; none changed. Section 9 gives the source paths and executable entry point. No participant analysis or task physics is run by this source-table check.

The visible follow-up chain is in `artifact/followup/interactions/`. Its index binds each call to its trial and segment. Exact prompts and response files retain their bytes; analysis request files contain the submitted code and public history, and tool-output files contain the archived JSON result. All visible agent-message text is supplied in order; the last message is the selected terminal response. Earlier visible messages are not forwarded as chat history: only the selected analysis program and its result enter the next request's `analysis_records`. The reader verifies this distinction from the cumulative prompts. This is 120 calls, not 177 calls or 177 trials.

Intent and receipt projections omit paths and account/thread identifiers. Raw provider streams, including hidden reasoning, remain local; their source hashes establish a binding but do not let readers reconstruct omitted fields. Fifty-three collector and dependency files across the three segments are inspection-only `.py.txt` files, with source and distributed hashes and explicit path-redaction status. They document collection without providing a runnable redispatch command. The reader's corruption controls reject missing calls, changed identities, wrong conditions, broken code/result/history links and altered finals. These are tests of the supplied evidence chain, not new scientific trials or execution of participant programs.

Full operational archives remain local. Hashes do not establish independent laboratory replication or immutable provider-model/tier identity. The proposed population, live query-selection and representation-transfer studies were not run; the finite-acquisition companion retains its separate, unchanged sample and claims.

J.1 History-level outcome accounting

Table 19 retains the four purposeful histories and their two repeats within each profile. It describes these assignments only; neither the cells nor their repeated contexts are independent new worlds. The reader derives this table and the main-text family, transition and rule-sensitivity tables from the bound minimal finals, allocation and menu census. Technical assignments retain no scientific label. All completed follow-up labels are P or L; there are no U or protocol outcomes.

Profile	History	Family	A: P/L/T	B: P/L/T	C: P/L/T
Astra	probe-01	Continuous	1/1/0	2/0/0	1/1/0
Astra	probe-02	Continuous	2/0/0	2/0/0	1/1/0
Astra	probe-03	Route	2/0/0	2/0/0	2/0/0
Astra	probe-04	Route	1/0/1	2/0/0	1/0/1
Sol	probe-01	Continuous	1/1/0	0/2/0	2/0/0
Sol	probe-02	Continuous	2/0/0	2/0/0	1/1/0
Sol	probe-03	Route	2/0/0	2/0/0	2/0/0
Sol	probe-04	Route	2/0/0	2/0/0	2/0/0

Table 19: All follow-up history/arm outcomes; T denotes technical loss with no scientific label. Every cell has two planned assignments. All completed labels are P or L. Repeated contexts are not independent draws of new worlds; P is continuation feasibility, not measured full-task completion.

K Supporting fixed-contract control

The comparison holds the archived tables, rule definitions, menu and task-label map fixed. It removes operations rather than withholding information from a baseline. We evaluate (i) native actions and rule equality, (ii) native actions and the single-distinct-row condition, and (iii) independent per-record evaluation of each supported rule. The last baseline never needs a partner's table. Only after producing its results do we compare them with paired cells.

Proposition (per-record sufficiency under a fixed contract). For fixed tables A_i, a common menu, and a fixed family $\mathcal{R}$ of deterministic rules with specified tie breaking, the collection $\{(i, r, r(A_i)) : r \in \mathcal{R}\}$ determines every supported-rule-family cell and every observed-rule cell whose rule belongs to $\mathcal{R}$. *Proof.* Each cell is one of those evaluations; no partner table is an argument of $r(A_i)$. A fixed action-label map also determines its label. This statement does not establish the validity of that map or identify an actor's internal computation.

Two weaker checks need fewer rule evaluations. If $r_1 = r_2$, then $c_{i1} = c_{i2} = a_i$. If a table has a single distinct row, mean and worst case select the same column. Duplicate-containing tables with other distinct rows do not satisfy this condition. The full-scan equality is thus a compositional identity of the contract. The empirical question is how often the weaker checks suffice and what contrasts remain; agreement with the full scan is not an independent discovery of the identity. We measure recoverable cells, not a wall-clock speedup or causal attribution accuracy.

Baseline information	Observed cells	Family cells
Native actions + identical observed rules	32/32	—
Native actions + one-distinct-row check	—	29/32
Independent two-rule scan per record	32/32	32/32
Additional cells from full table after scan	0	0

Table 20: Mandatory condition only. Counts describe recoverable displayed cells, not independent observations or accuracy on external diagnoses.

L Prospective choice-prediction protocol and source access

History construction and target branches. The preparation fixed two instances per disclosed task family with seed 2026103001, before model outputs. The purpose was a small feasibility panel with fresh histories and legal continuation opportunities, rather than a population sample or a balanced difficulty comparison. In continuous control, a privileged preparation controller generated six transitions by choosing the menu action with the smallest next-state norm. In route tasks, the fixed action prefix was $(0, 0), (1, 0), (2, 0), (0, 0)$; initial fuel was set to retain a known feasible continuation at the source and both prescribed branches,

with a three-unit margin. Those preparation controllers are not participant-model behavior. All twelve source/target packets passed the precollection continuation check; the continuous preparation would stop on a failed witness rather than resample. The four histories differ from the earlier published histories. The original construction code and generation record are supplied for inspection in `artifact/prospective-design/`; the reader does not execute them.

Histories	Family	Source → target step	Two separate branch actions
h01, h02	Continuous	$6 \to 7$	$(0.6, 0)$; $(-0.6, 0)$
h03, h04	Route	$4 \to 5$	$(0, 0)$; $(2, 0)$: advance 1; 3 nodes

Table 21: Fixed source–target construction. Step counts completed transitions. Each target preserves the entire source history and adds one transition and its observation under the indicated exogenous action. The two branches are alternatives from the same source, not consecutive transitions. Branch actions were fixed before source-model outputs and need not equal the source proposal.

The model actors receive the generated public histories, not the privileged preparation state or continuation witnesses. Each target contains the full source prefix and one additional action–observation transition. The target branches and source-proposed action are separate quantities.

Allocation and information boundaries. The pilot fixes four new purposeful histories in the two existing disclosed families, two profiles, one source per history/profile and two precommitted target branches per source: eight sources, 32 observers and 16 targets. Eight initial references make 64 assigned roles. Each target branch is a precommitted branch from the source history; it need not follow the model's source-proposed action. Both observers receive that fact. Every role uses a fresh context; the source never receives future target snapshots. The source's private analysis transcript is not replayed to observers or target, although its information can be summarized in the record. The four prose fields and numeric table are passed only to the with-record observer. The target receives neither record nor observer responses. Exact role instructions, schemas, packets, allocation, prompts and selected terminal responses are supplied.

The source final schema requires the action, numeric table, declared rule and four prose fields; the target final schema requires the action alone. Both observer arms receive both schemas and complete actor instructions. This evaluates prediction across these output protocols; a same-protocol target arm and an isolated protocol-change effect are outside the allocation. Source and target actors may request up to three local analyses, followed by a final choice; observers make one request without an analysis tool. Both observer arms have the same requested model/max/Fast profile and 900-second request allowance. Actor allowance is 3,780 seconds, each local analysis 30 seconds. These allowances do not require equal realized token use, and the added record increases input length. Requested profiles and fresh local contexts are recorded; immutable remote model identity is not attested.

Reference gate and completion. Two hand-authored policy cases are evaluated with and without a record for each profile. The initial gate requires eight valid transport/protocol results and all four with-record predictions correct. Astra's with/without accuracy is 2/2 and 2/2; Sol's is 2/2 and 1/2. Without-record accuracy is descriptive, not a gate. The later Sol completion repeats these four references: all valid, with-record 2/2 and without-record 1/2. Only then are the two remaining predictions locked and the target requested. Reference success does not establish scientific prediction benefit or durable provider availability.

The original segment has 61 completed, one interrupted and two unstarted roles. The new segment has seven completed roles, including four new controls and the three remaining scientific roles. Their union covers all 64 original assignments, but the 68 completed segment roles are not 68 independent samples. The interrupted original request is neither deleted nor marked successful. Completion was decided after partial-result inspection and adds no new history. It restores the planned target denominator without transforming this pilot into a confirmatory sample. The original partial panel remains visible.

Scoring and structural diagnostics. For each valid target, an observer scores one iff its action ID equals the target's menu-bound ID. Invalid observer protocol would score zero; unavailable source or target leaves

the paired difference in $[-1, 1]$. The primary descriptive quantity averages the two target differences within each history, then the four histories, separately by profile. There are no protocol losses in the completed original or completion results. We report assigned, started, completed, interrupted and unstarted roles separately. Empty denominators would be unavailable, never success. There is no pooled primary estimate, population significance test or model ordering.

The reader derives source validity, declared selection, both supported-rule choices, row count, distinct-row count and proposal match directly from the eight exact finals. It checks all menu/action identities. All eight source records match; seven are single-row and one Sol record has two distinct rows. All eight select the same action under both rules, forced in the seven single-row cases and observed in the remaining one. No row duplication occurs. The source-action copy baseline is legal for every target. The numeric table plus prose is one treatment. The semantic informativeness of the prose was not independently evaluated, and the seven single-row tables do not establish that the prose is uninformative. Table 13 reports joint correctness for the combined panel; equal marginal counts can include different correctly predicted targets. Identical-input target variability and task-specific informative-record sensitivity were not separately measured. The family and history breakdowns in Tables 14 and 22 were added after inspecting the combined results. They preserve every assigned history and target and leave the prespecified aggregate unchanged.

Profile	History	Family	With record	Without record	Net correct
Astra	h01	Continuous	1/2	2/2	−1
Astra	h02	Continuous	1/2	1/2	0
Astra	h03	Route	2/2	2/2	0
Astra	h04	Route	2/2	2/2	0
Sol	h01	Continuous	2/2	1/2	+1
Sol	h02	Continuous	1/2	1/2	0
Sol	h03	Route	0/2	0/2	0
Sol	h04	Route	0/2	1/2	−1

Table 22: Post hoc breakdown by the eight source-history/profile units. Each row contains its two linked target branches. Sol h04 includes the separately completed final target; the original fifteen-target panel remains in Table 12. Histories are not repeated independent draws.

Usage and reserves. The initial segment has 96 requests: 95 measured and one unknown Sol request. Astra's 47 measured requests used 747,683 input and 282,909 output tokens, including 150,016 cached input tokens; Sol's 48 measured requests used 854,673 input and 345,366 output tokens, including 84,480 cached input tokens. The new segment uses seven Sol requests: four controls, two observers and one target, with 88,601 input and 34,066 output tokens, including 32,000 cached input tokens. Cache is a subset of input, and reasoning is a subset of output; neither is counted again. The new segment contains no analysis request.

Frozen local credit equivalents are 1,013.60680 for initial Astra, 250.12470 for initial Sol, and 22.85310 for the Sol completion. The original unknown request still reserves 1,050,000 input plus 128,000 output tokens, equivalent to 169 credits. Across segments: 103 issued requests, 102 measured, one unknown; measured equivalents 1,286.58460, reserve-inclusive 1,455.58460. The initial 10,000 local bound remains; the separately approved completion bound is 1,690 with at most ten new requests. Neither is a provider invoice cap. Analyst use and older follow-up reserves are outside this accounting. All 102 measured receipts have known cache splits; this does not make the unresolved request's usage known. No 429 appears in these receipts; the original interruption's cause remains undetermined.

Prospective evidence and check limits. The prospective evidence retains its own exact manifest within the single supplementary archive. It includes 103 exact prompts and their schemas, 102 selected terminal responses, 34 analysis program/result pairs, eight source records, and projected request receipts and journal chronology. Analysis code is readable data; the reader never executes it or a physical simulator. The main package binds the supplement's manifest. Its source reader reconstructs prompts, re-parses source tables, checks source–prediction–target order, and recomputes initial and combined panels and usage. The implementation shares the collector's parser and scalar readout; negative controls challenge its evidence

links. Local source hashes and projected timestamps are not independent provider attestation. Raw account identifiers, hidden reasoning and provider-added context are not supplied. The reader audits the supplied evidence, not unobserved internal computation or general-world prediction.

M Evidence access and operational accounting

Table 23 specifies access separately for each evidence set. The bundle contains LaTeX sources, disclosed current-study mechanisms, older numerical replay and the follow-up source reader. The older panel's 80 payloads reconstruct prompt hashes; archived participant programs are never executed. The unchanged companion is optional supporting material. Its catalog, ordered draw, 5,000 histories, queries, certificates and replay are in a separate companion bundle; they are not needed for the main-paper checks. Existing-input replay is not new data collection or historical process attestation.

For the follow-up, the standard-library reader parses exact numeric lexemes from 46 minimal finals, including 31 tables. It derives validity, both rule choices and row diagnostics, compares archived decisions, verifies menu and profile identities, and checks 46 labels by lookup in the bound census. A separate interaction reader now reconstructs all 120 archived application prompts from the condition instructions, four fixed public packets and the exact cumulative analysis prefix. It links 118 terminal response files, all 177 visible agent messages and 72 analysis input/output pairs to the same allocations and finals. Two failed requests retain no response and unknown usage. Collector sources are supplied as inspection-only text; participant programs and collector code are never executed by these checks. Intent and receipt files are explicit field projections with original source hashes. This permits an audit of the supplied application-visible chain, not independent attestation of historical execution, provider-added context or hidden reasoning. Original streams with machine/account identifiers remain local. The earlier revision added numerical re-evaluation of the existing pilot sensitivity actions. The paper also reports the prospective choice-prediction pilot and its separately recorded completion. The prospective supplement contains all eight exact source records, 32 scientific predictions, 16 target finals and 12 reference finals; 103 exact prompts, 102 terminal response files and 34 analysis code/result pairs support reconstruction of the visible inputs. The unresolved request has a prompt and an unknown-usage reserve. Its reader checks action/menu identities and chronology and recomputes both result panels without executing submitted programs or physics. It shares the collector's record parser and scalar calculation, so this is reproducibility, not an independent implementation. The prospective component is bound by hash to the main component. Both are restored from the single supplementary archive.

The single supplementary archive stores repeated UTF-8 lines once in readable JSON shards, with an indexed path-to-line mapping and the original file hashes. PDF files remain directly readable. The extraction utility rejects unexpected paths, invalid line references and altered bytes before restoring the two complete components. Component manifests and the main-to-prospective binding are then verified by the evidence reader. This representation changes storage only; it preserves all source files and both execution segments. To obtain the original directory layout for direct inspection, run:

```
python3.12 -I -B -S verify.py --extract-to ../ruu-evidence
```

From `../ruu-evidence/paper-package`, the component reader is:

```
python3.12 -I -B -S verify_all.py --prospective-dir ../prospective
```

The additional scalar check derives the A2 strict-dominance comparisons, the four-cell paired table, and the family and history breakdowns directly from the supplied numeric source and terminal responses. It also checks the fixed source–target prefixes and branch actions. It does not execute participant programs or the history generator.

Evidence	Supplied material	Boundary of the check
Prospective supplement	Eight exact source records; 103 prompts, 102 terminal responses, 34 analysis pairs; allocation, menus, instructions and projected receipts	Recompute original/combined predictions and record structure; retain interrupted attempt; projected chronology is not provider attestation
Follow-up sources	46 minimal finals, including 31 numeric tables and rules; allocation, common/A/B/C instructions, schema, menu and bound labels	Recompute final decisions and labels by lookup; no provider or historical-runtime attestation
Follow-up interaction	120 prompts, 118 terminal responses, 177 visible messages, 72 analysis input/output pairs; inspection-only collector sources	Reconstruct cumulative application prompts and bind finals; projected receipts do not independently authenticate historical execution or provider context
Older common-menu panel	All 80 request payloads, prompt renderer, instructions, histories and label-replay inputs	Reconstruct archived prompts and hashes; participant programs are inspected as data, never executed
Pilot selector sensitivity	Twelve original action grades with full witnesses and certificates, action lock, fixed plan inputs and reader	Recompute plans, selectors and conditional labels; compare bound evidence for every action
Mechanisms and labels	Disclosed current-study mechanisms and older numerical replay sources	Conditional continuous premises remain open; replay is not independent historical or laboratory attestation

Table 23: Reader access by evidence set. Final-table recomputation, visible interaction consistency and historical attestation are separate capabilities. No hidden model reasoning is included.